

EXPERIMENT 3

Survey Report on Emerging Trends in Low-Code Application Development

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A survey of current developments, adoption drivers, emerging technologies, enterprise practices, challenges, and future directions in low-code application development.

Academic Report

1. Abstract

Low-code application development has evolved from a productivity technique into a broader application delivery approach used by professional developers, business technologists, and citizen developers. Current platforms increasingly combine visual development with artificial intelligence, automation, reusable components, enterprise integration, governance, and observability. Recent industry research identifies agentic AI, AI-assisted development, citizen development, composable architectures, legacy modernization, and stronger governance as important directions shaping the low-code market.

2. Introduction

Low-code development uses visual interfaces, reusable components, workflow builders, preconfigured services, and limited hand-written code to accelerate software creation. The approach reduces the amount of repetitive programming required for many business applications while still allowing professional developers to extend solutions when custom logic is needed.

The current generation of low-code platforms is increasingly connected with generative AI and AI agents. Microsoft describes an AI-first direction in which natural-language interaction, Copilot capabilities, applications, workflows, and agents become part of a common development environment. Gartner's 2025 enterprise low-code research also highlights AI-assisted tooling, composable architectures, and built-in governance as major characteristics of modern enterprise low-code application platforms.

3. Objectives of the Survey

- Identify major emerging trends in low-code application development.
- Examine how AI and automation are changing low-code platforms.
- Understand enterprise adoption, integration, and modernization use cases.
- Study governance, security, scalability, and other challenges.
- Identify likely future directions for low-code development.

4. Emerging Trends in Low-Code Application Development

4.1 AI-Assisted and Natural-Language Development

Generative AI is becoming a core capability of low-code platforms. Instead of configuring every component manually, users can increasingly describe application requirements in natural language and receive assistance with application structure, formulas, workflows, data operations, and user interfaces. Microsoft identifies the combination of AI tools, Copilot capabilities, and low-code development as part of an AI-first development direction.

4.2 Agentic AI and AI-Powered Applications

A major emerging trend is the integration of AI agents into applications and development platforms. Agents can perform multi-step tasks, interact with tools, retrieve information, and support business workflows. Gartner's 2026 research specifically addresses governance of AI-agent development on low-code application platforms, showing that agent development is becoming an important enterprise concern.

4.3 Citizen Development

Low-code platforms continue to expand software creation beyond traditional software engineering teams. Business users can create forms, dashboards, workflows, automations, and departmental applications with limited programming. Gartner identifies citizen development as one of the drivers expected to support low-code market growth.

4.4 Composable and Reusable Architecture

Modern low-code platforms increasingly emphasize reusable modules, APIs, connectors, services, and components. This supports composable application design, where larger solutions are assembled from smaller reusable building blocks. Such approaches can reduce duplication and make application development more maintainable.

4.5 Legacy Modernization

Organizations are using low-code platforms to modernize older systems and create new user experiences around existing enterprise data and services. Low-code can provide a faster layer for building new applications while connecting to established databases, APIs, and business systems.

4.6 Governance, Security and Compliance

As low-code adoption expands, governance becomes increasingly important. Organizations need controls for identity, permissions, application lifecycle management, data access, compliance, monitoring, and secure use of AI-generated functionality. Gartner's research emphasizes governance as a key requirement for enterprise low-code environments and for AI-agent development.

5. Survey Findings from Recent Industry Research

Source / Year	Key finding relevant to low-code
Gartner, 2025	Low-code platforms are being shaped by AI-assisted tooling, composable architectures and built-in governance.
Gartner, Nov. 2025	Low-code development technologies are forecast to reach \$58.2 billion by 2029; agentic AI and citizen development
Microsoft, 2025	AI-first development is bringing natural-language assistance, Copilot capabilities, applications, workflows and agents
OutSystems, 2025	Research involving nearly 1,700 IT professionals highlights generative AI, low-code development and digital transfor
App Builder, 2025	Improving developer productivity was reported as the leading motivation for low-code adoption in its survey.
Gartner, 2026	Organizations developing AI agents on low-code platforms need stronger governance practices before broad produc

6. Major Benefits of the Emerging Approach

- **Faster development:** visual components and reusable services reduce repetitive implementation work.
- **Higher productivity:** developers can focus more on business logic and complex engineering tasks.
- **Broader participation:** business teams can contribute to application creation.
- **Rapid prototyping:** ideas can be converted into working applications and workflows quickly.
- **Better integration:** connectors and APIs help applications interact with existing enterprise systems.
- **AI-enabled experiences:** natural-language assistance and agents can be embedded into business applications.

7. Challenges and Considerations

Despite its benefits, low-code development does not remove the need for software engineering discipline. Organizations must manage application sprawl, security, data governance, integration complexity, vendor dependence, performance requirements, and lifecycle management.

AI introduces additional considerations. AI-generated application logic and agent behavior need testing, monitoring, access controls, human oversight, and clear accountability. Gartner's recent research emphasizes that governance is particularly important as organizations move AI agents from experimentation toward production.

8. Future Scope

The future of low-code development is likely to be increasingly AI-assisted and agent-enabled. Development platforms are expected to move toward conversational application creation, intelligent workflow automation, reusable components, stronger integration with enterprise data, and more automated testing and deployment. Professional developers and business users are likely to work together in hybrid development models rather than low-code replacing conventional programming.

9. Conclusion

The survey shows that low-code application development is progressing from simple visual app building toward an AI-enhanced software delivery model. AI-assisted development, agentic AI, citizen development, composable architecture, legacy modernization, and enterprise governance are among the most significant emerging trends. The long-term value of low-code will depend not only on development speed but also on security, scalability, maintainability, integration, and responsible governance.

10. References

1. Gartner, *Gartner Magic Quadrant for Enterprise Low-Code Application Platforms*, 2025.
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3. Gartner, *How to Govern AI Agent Development on Low-Code Application Platforms*, 2026.
4. Microsoft Power Platform, *Identify your readiness for AI-first development with agents and Microsoft Power Platform*, 2025.
5. OutSystems, *The 2025 State of Application Development*, 2025.
6. App Builder, *Survey Report: 2025 Trends in App Development*, 2025.